PL-300 study guide: Model the data

Data modelling, DAX and model performance

16 exam skills · 16 worked examples
Original practice questions with explained answers

Skills measured as of 20 April 2026 · Exam scope checked 11 November 2026

How to study each skill before exam practice

Read the definition and worked example, then diagnose the mistake. Answer the question before opening the answer page. Explain why each alternative fails.

Definition	Example	Common mistake	Question	Answer

Model the data is 25–30% of PL-300. Data checks use course CSVs. DAX and menu routes still need product rehearsal.

The 16 exam skills and their starting pages

1.1 Configure table and column properties · 6

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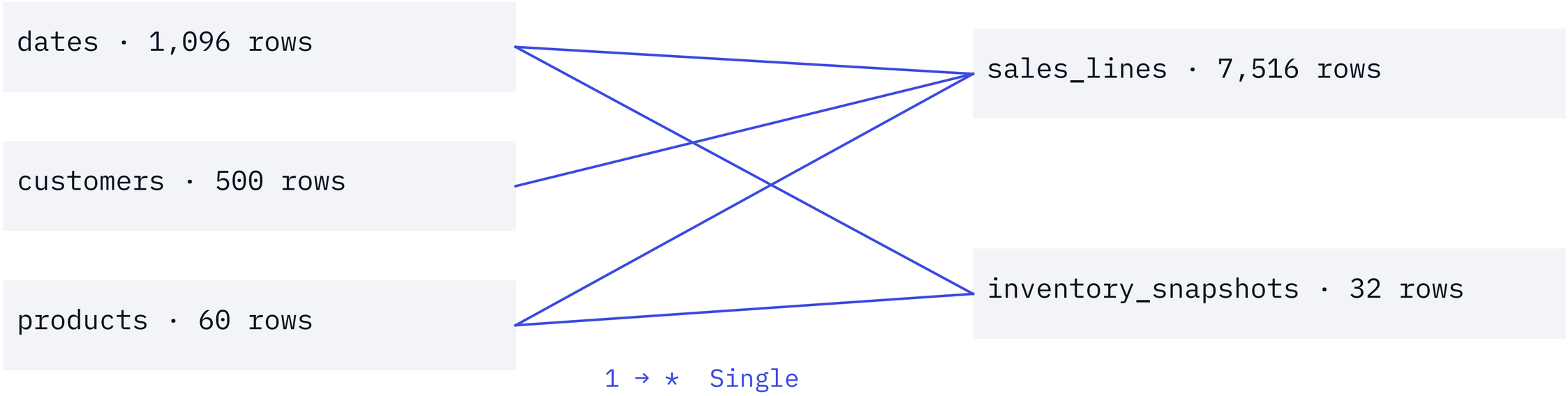
3.1 Identify and remove unnecessary rows and columns · 77

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Nakheel’s model supplies the guide’s worked examples

A fact table stores events or snapshots. Dimensions describe them. Sales lines record purchases. The new stock fact records one product on one Friday. Dimensions filter both facts independently.



Use Single dimension-to-fact filters. Order date is active, delivery date inactive. The stock fact relates to dates and products.

Chapter 1: design and implement a data model

Field properties, relationship roles and calculation choices

Design the model

Write calculations

Improve performance

Table and column properties control how fields appear

A property describes a field to Power BI. Formatting changes its display. A data type changes how values are stored and compared.

Field property	dates[year]	What it controls
Data type	Whole number	Stored representation
Format	0	Display of values
Summarize by	Don’t summarize	Default aggregation
Data category	Uncategorised	Meaning of the field
Sort by column	year	Ordering of labels
Hidden	No	Visibility to report authors
Display folder	Calendar	Field organisation
Synonyms	calendar year	Natural-language names

Years and identifiers usually need Don’t summarize

An unfiltered card sums all 1,096 year values to 2,219,399. A year axis should instead show 2024, 2025 and 2026 as labels.

Sum of year

2,219,399

Year as a label

2024 2025 2026

A year label identifies a period. Adding the year numbers answers no useful question here.

Month names need a numeric sort column

Set dates[month] to Sort by column → dates[month_no]. Each month name must map to one month number. A data category such as City helps identify a location.

Alphabetical order	Calendar order	Sort key
April	January	1
August	February	2
December	March	3
February	April	4

Sort the month name by its month number. Clean city spellings before setting City as a data category.

Limits and common mistakes: configure table and column properties

When another option fits

Hide relationship keys when report authors do not need them. Hidden columns still occupy model memory and remain queryable. Use security rules to restrict data.

The mistake to diagnose

City is a geographic label, not a cleaning rule. The file contains Amman and amman, plus Zarqa and Al Zarqa. Standardise the values in Power Query before mapping.

Exam practice: configure table and column properties

A chart lists April, August and December before February. Revenue is correct. How should you make the months follow calendar order?

A Sort the visual by revenue.

B Change month’s data type to Date.

C Set month’s Sort by column to month_no.

D Set month’s Summarize by to Don’t summarize.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: C · configure table and column properties

A Revenue order changes with the values. It does not establish calendar order.

B Names such as April do not identify complete dates. Keep the month label as text.

C The numeric month number gives each text label its calendar position.

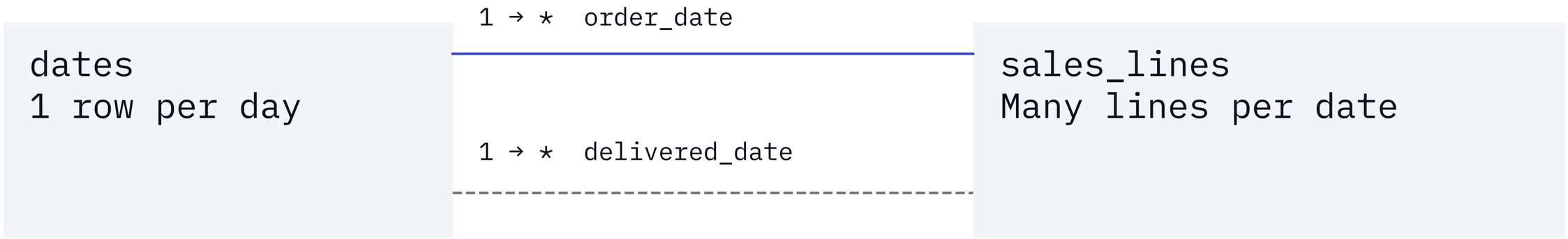
D This controls aggregation. It does not define the order of text labels.

Full worked sequence: D18 pp. 47, 82–85 · Lab D18 / 04

Microsoft Learn · properties

A date dimension can describe order dates or delivery dates

A role-playing dimension describes the same business entity in different roles. Nakheel uses dates for the day an order was placed and the day it was delivered.



Solid = active order-date role. Dashed = inactive delivery-date role used by USERELATIONSHIP.

Two date roles can use a measure or separate dimensions

With one calendar, USERELATIONSHIP gives April delivery revenue of 17,815.50, versus 20,237.00 by order date. For both roles at once, use separate dates and Delivery Dates tables for the matrix below.

```
Delivered Revenue =
CALCULATE(
[Completed Revenue],
USERELATIONSHIP(dates[date], sales_lines[delivered_date])
)
```

dates: order month	Delivery Dates: Apr	Delivery Dates: May
Apr 2026	17,524.50	2,712.50

Two independent date tables support both axes. The April order row sums to 20,237.00.

Limits and common mistakes: implement role-playing dimensions

When another option fits

Use a second Delivery Dates dimension when users must filter both roles independently. One alternate-date measure can use USERELATIONSHIP without adding another table.

The mistake to diagnose

Changing the active relationship changes the default date role for other measures. Both direction does not create independent order-month and delivery-month filters.

Exam practice: implement role-playing dimensions

A matrix needs order month on Rows and delivery month on Columns at the same time. Which design supports independent filtering by both roles?

A Keep one dates table and use only USERELATIONSHIP.

B Create Delivery Dates and actively relate it to delivered_date.

C Make the delivery relationship active and remove the order relationship.

D Set both date relationships to bidirectional.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · implement role-playing dimensions

A A single date table cannot supply two independent sets of date filters.

B Separate date dimensions can filter order date and delivery date independently.

C This loses the default order-date path needed for the row filter.

D Direction controls filter propagation. It does not give one dimension independent roles.

Full worked sequence: D18 pp. 74–81 · Lab D18 / 04; D20 / 06 step 2

Microsoft Learn · roles

Cardinality describes keys and direction controls filters

Cardinality says whether a relationship key is unique on each side. Cross-filter direction says which table can filter the other. A dimension usually filters many fact rows.

Upstream orders	sales_lines: one row per line	Cardinality
100003	100003 · line 1	1 : *
	100003 · line 2	
	100003 · line 3	

Upstream orders is shown for comparison. The supplied model already includes order fields in sales_lines.

Counting customers in the fact table follows category filters

For the all-status Outerwear category: COUNT(customers[customer_id]) gives 500 with Single and 319 with Both. A fact DISTINCTCOUNT also gives 319. Every Outerwear customer key in these files has a matching customer row.

```
Customers with sales =
DISTINCTCOUNT(sales_lines[customer_id])
```



All-status Outerwear: both counts are 319 when filters reach the counted table. These Outerwear rows have no unmatched customer IDs.

Limits and common mistakes: define cardinality and cross-filter direction

When another option fits

Use Both only when the required filter propagation justifies it. CROSSFILTER can change the relationship's filter direction for one calculation. Many-to-many can be valid, but duplicate dimension keys need investigation.

The mistake to diagnose

If a fact key has no matching dimension row, Power BI can create a blank dimension member. A nonblank dimension-ID count can then differ from the fact-ID count. Direction changes do not repair unmatched keys.

Exam practice: define cardinality and cross-filter direction

The model has unique customer IDs in customers, Single direction to sales_lines, and category filters from products. Every category shows 500. You need distinct customer IDs present in each category's sales lines. What should you do?

A Change all relationships to Both.

B Use DISTINCTCOUNT(sales_lines[customer_id]).

C Change the customer relationship to many-to-many.

D Mark customers as a date table.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · define cardinality and cross-filter direction

A	This changes filter propagation throughout the model and is unnecessary for the stated count.
B	Category filters reach sales_lines, where the requested IDs can be counted.
C	Changing cardinality does not fix the location of the count.
D	Date-table validation has no role in counting customers.

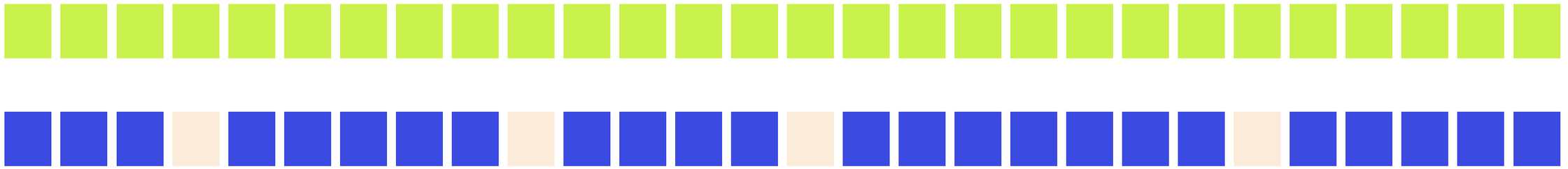
Full worked sequence: D18 pp. 53–68 · Lab D18 / 03 and optional 06

Microsoft Learn · relationships

A common date table supplies every day in the period

A date dimension has one unique, nonblank date per row. Classic time intelligence needs a continuous date range covering full years. Mark the date column for this course's classic DAX route.

Calendar: 2024–2026 · 1,096 consecutive dates



Fact strip: schematic gaps · 714 of 730 sales-period days have orders

Use the calendar for periods with no sales. The full date table also covers dates outside the sales interval.

CALENDAR supplies a complete three-year date range

The supplied dates table has 1,096 rows for 2024–2026. Set the date data type, mark dates[date], sort month by month_no and use the common date fields in visuals.

```
Dates Example =
CALENDAR(DATE(2024, 1, 1), DATE(2026, 12, 31))
```

Calendar rows

1,096

April 2026 YTD

62,497.50

Limits and common mistakes: create a common date table

When another option fits

CALENDARAUTO derives its range from eligible model dates, so unrelated historical dates can extend it. Power Query can also build the date table. Disable Auto date/time when using the shared table.

The mistake to diagnose

The fact has 714 distinct order dates across a 730-day sales interval. Missing sales days belong in the calendar. Merely marking a table does not repair gaps or an inactive relationship.

Exam practice: create a common date table

Classic time intelligence uses a date column copied from distinct sales dates. Some days have no sales. Which change gives the model a suitable shared calendar?

A Use only the dates that have revenue.

B Add a text month field to the fact table.

C Create continuous full-year dates, mark the date column and relate it to sales.

D Set the date relationship to Both.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: C · create a common date table

A This keeps the gaps that the calendar must fill.

B A text label does not supply a daily date dimension.

C This supplies the continuous date range and a filter path to the sales fact.

D Direction does not add missing dates or validate the date column.

Full worked sequence: D18 pp. 42–50; D20 pp. 8–17 · Lab D18 / 02; D20 / 02

Microsoft Learn · dates

Stored calculations and measures answer different questions

A calculated column stores a result for each row. A calculated table stores rows from a DAX expression. Import models compute these at refresh. A measure evaluates under the report's filters.

Calculation	Evaluated	Stored in Import	Responds to slicers
Power Query column	Before load	Yes	No
DAX column	Refresh	Yes	No
DAX table	Refresh	Yes	No
Model measure	Query time	No result storage	Yes
Visual calculation	Visual evaluation	No model result	Within the visual

A category summary stores the refresh-time revenue

Outerwear’s stored category revenue is 76,637.00. Selecting online leaves it unchanged. A measure in the original star responds and returns 25,315.50.



A measure re-evaluates under the channel filter. A stored summary’s numeric column retains its refresh-time values.

Limits and common mistakes: identify uses for calculated columns and tables

When another option fits

Prefer source or Power Query columns for transformations that belong before load. Use a model measure for reusable values that follow slicers. Visual calculations apply only inside their visual.

The mistake to diagnose

A row-level percentage stored in a column does not automatically make a valid total percentage. Calculate the ratio of totals in a measure when the report needs a filtered ratio.

Exam practice: identify uses for calculated columns and tables

A revenue percentage must recalculate for slicers and be reused across several report pages. Which calculation type fits?

A A DAX calculated column.

B A DAX calculated table.

C A model measure.

D A visual calculation in one chart.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: C · identify uses for calculated columns and tables

A Stored row values do not recalculate under report slicers.

B A stored table is evaluated at refresh, not on each slicer change.

C The measure recalculates under the report’s filters and can be reused across visuals.

D It belongs to one visual, so it does not meet the reuse requirement.

Full worked sequence: D19 pp. 11–13; D20 pp. 67–70 · Lab D20 / 03 and 06 step 1

Microsoft Learn · locations

Chapter 2: create model calculations by using DAX

Aggregation, filter context, time periods and reusable calculations

Design the model

Write calculations

Improve performance

An aggregation summarises the rows visible to a measure

An aggregation returns a single value from many values. SUM totals amounts, COUNTROWS counts rows and DISTINCTCOUNT counts different values in a column. Filters determine the rows included.

Order 100003 · SUM	COUNTROWS	DISTINCTCOUNT order_id
47.00	**3**	**1**

The same three sales lines contribute one distinct order.

Revenue, orders and customers use different aggregations

With no filters: 340,370.00 revenue, 3,000 orders and 462 fact customer IDs. Completed only: 316,023.50 revenue, 2,771 orders and 452 customer IDs across 6,951 lines.

```
Total Revenue = SUM(sales_lines[line_value])
Orders = DISTINCTCOUNT(sales_lines[order_id])
Customers = DISTINCTCOUNT(sales_lines[customer_id])
Average Order Value = DIVIDE([Total Revenue], [Orders])
```

Completed revenue	Completed orders	Average order value
316,023.50	**2,771**	**114.05**

Limits and common mistakes: create single aggregation measures

When another option fits

AVERAGE(line_value) answers average line value. For average order value, divide revenue by distinct orders. Explicit measures give a reusable name and definition to the calculation.

The mistake to diagnose

COUNT skips blanks. COUNTROWS counts rows. DISTINCTCOUNT includes a blank as one distinct value. Choose the function from what one observation means.

Exam practice: create single aggregation measures

Completed revenue is 316,023.50 across 6,951 lines in 2,771 orders. A card shows 45.46 as average order value. Which formula answers the order question?

A AVERAGE(sales_lines[line_value]).

B DIVIDE([Completed Revenue], CALCULATE([Orders], sales_lines[status] = "completed")).

C COUNTROWS(sales_lines).

D SUM(sales_lines[price]).

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · create single aggregation measures

A This returns the average line value, 45.46.

B Revenue divided by completed distinct orders gives 114.05.

C This counts lines and returns no average.

D Unit prices alone do not account for quantities or orders.

Full worked sequence: D19 pp. 15–28 · Lab D19 / 02

Microsoft Learn · aggregations

CALCULATE evaluates an expression with changed filters

Filter context is the set of filters applied to a calculation. CALCULATE evaluates an expression after changing that context. A filter argument usually replaces a filter on the same column.

Outerwear row	Before CALCULATE	After status = completed
products[category]	Outerwear	Outerwear
sales_lines[status]	cancelled	completed
sales_lines[channel]	online	online

CALCULATE replaces the same-column status filter and keeps the category and channel filters.

A completed-status measure replaces the status filter

Across all dates and products, a matrix lists completed 316,023.50, cancelled 21,998.50 and pending 2,348.00. Completed Revenue returns 316,023.50 on every status row, while keeping filters on other columns.

```
Completed Revenue =
CALCULATE([Total Revenue], sales_lines[status] = "completed")
```

Status row	Total Revenue	Completed Revenue
completed	316,023.50	316,023.50
cancelled	21,998.50	316,023.50
pending	2,348.00	316,023.50

REMOVEFILTERS and KEEPFILTERS change different parts of context

Outerwear’s share is 76,637.00 / 316,023.50 = 24.3%. REMOVEFILTERS(products) clears product filters in the denominator. KEEPFILTERS keeps only rows that satisfy both the new status condition and the existing condition.



REMOVEFILTERS(products) removes category and product-name filters. Outerwear’s share is 24.3%.

Limits and common mistakes: use the CALCULATE function

When another option fits

For a simple condition, prefer a Boolean column filter. Use FILTER when you need a row-by-row table condition. CALCULATE also turns a current row into filters during context transition.

The mistake to diagnose

A Shoes filter replaces the category row filter, so Shoes revenue can appear on every category row. REMOVEFILTERS(sales_lines) can remove far more context than a category denominator needs.

Exam practice: use the CALCULATE function

Every category shows 100% in a share measure. The numerator and denominator both use Completed Revenue under the category filter. Which denominator preserves channel and date selections but removes product filters?

A [Completed Revenue].

B CALCULATE([Completed Revenue], REMOVEFILTERS(products)).

C CALCULATE([Completed Revenue], REMOVEFILTERS(dates)).

D SUM(products[list_price]).

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · use the CALCULATE function

A It keeps the category filter, so the numerator equals the denominator.

B It clears product filters while channel and date filters remain.

C It clears dates and leaves the category filter in place.

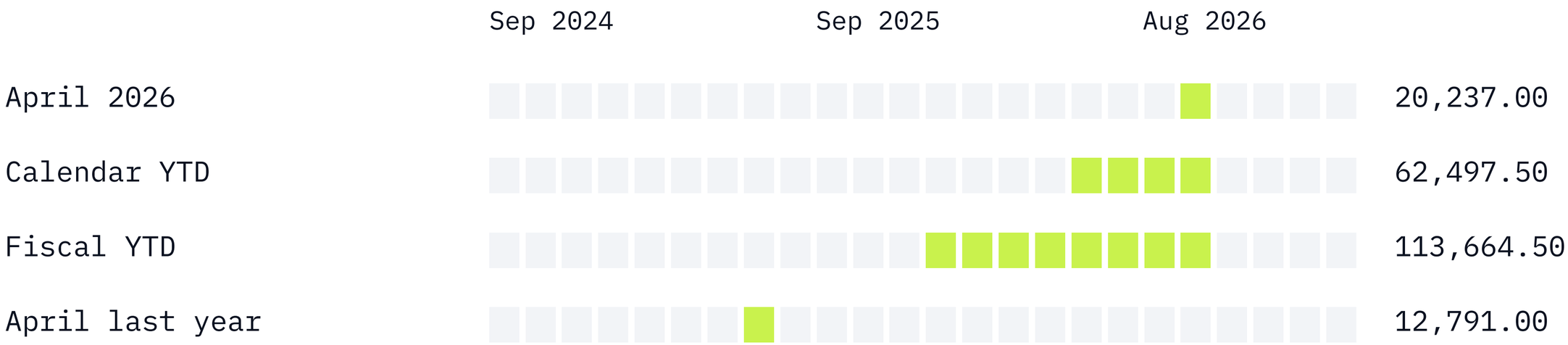
D Product list prices do not represent the revenue denominator.

Full worked sequence: D19 pp. 38–64; D20 pp. 38–40, 66 · Lab D19 / 03–05

Microsoft Learn · calculate

Time intelligence changes the dates used by a measure

Time intelligence compares or accumulates a measure over date periods. Use dates[date] from the common calendar for the classic functions in this guide. YTD starts on the first day of the year.



Each small mark is one month. The highlighted months filter the same completed-revenue measure.

April revenue changes with the selected date window

April 2026 completed revenue is 20,237.00. Calendar YTD is 62,497.50. April last year is 12,791.00. Fiscal YTD from September is 113,664.50. April YoY is +58.2%.

```
Revenue YTD = TOTALYTD([Completed Revenue], dates[date])
Revenue LY = CALCULATE(
[Completed Revenue], SAMEPERIODLASTYEAR(dates[date]))
YoY % = DIVIDE([Completed Revenue] - [Revenue LY], [Revenue LY])
```

April	YTD	April last year
20,237.00	**62,497.50**	**12,791.00**

Fiscal YTD needs the correct year boundary

Nakheel's fiscal year starts on 1 September. TOTALYTD can use an August year-end string, interpreted in the model's locale. The fiscal_year column gives a route without a date string.

```
Revenue FYTD =
VAR LastDay = MAX(dates[date])
VAR ThisFY = MAX(dates[fiscal_year])
RETURN CALCULATE([Completed Revenue],
REMOVEFILTERS(dates), dates[fiscal_year] = ThisFY,
dates[date] <= LastDay)
```

US-English locale: TOTALYTD([Completed Revenue], dates[date], "8/31")

In April, ThisFY is 2025-26. Clearing calendar filters allows the measure to include September through April.

Limits and common mistakes: implement time intelligence measures

When another option fits

Compare matching periods when the current year is incomplete. In this data, all 2026 against all 2025 gives –16.9%, while January–August against January–August gives +27.1%.

The mistake to diagnose

The calendar extends beyond the final sale. A blank current month compared with prior-year revenue can appear as –100%. Use an explicit missing-data rule before interpreting that as a business decline.

Exam practice: implement time intelligence measures

Nakheel’s year begins on 1 September. For April 2026, which period belongs in fiscal YTD?

A 1 January–30 April 2026.

B 1 September 2025–30 April 2026.

C 1 April–30 April 2025.

D 1 September 2026–31 August 2027.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · implement time intelligence measures

A This is calendar YTD, 62,497.50.

B This is the current fiscal year up to April, 113,664.50.

C This is April last year, 12,791.00.

D This is a future fiscal year, not the April 2026 accumulation.

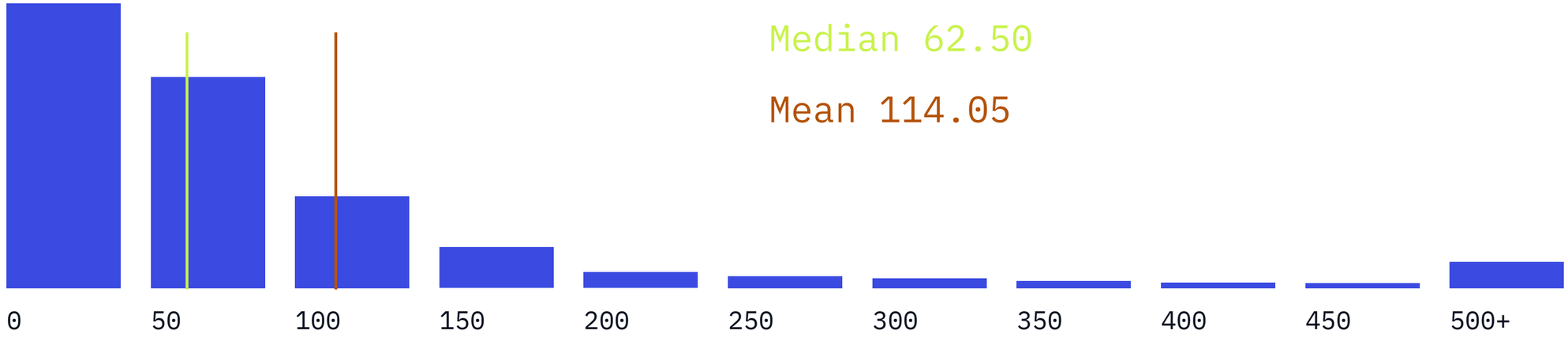
Full worked sequence: D20 pp. 8–29, 63–64 · Lab D20 / 02 and 06 step 2

Microsoft Learn · time

Mean and median describe the distribution differently

The mean divides the sum by the observation count. The median is the middle ordered observation. Large orders raise the mean more than the median. Define the observation before choosing a function.

Completed orders per 50-unit band; final band includes 500 and above



The median is 62.50. A long upper tail raises the mean to 114.05. Horizontal axis: completed order value.

Statistics over orders need one value per order

Across completed orders: mean 114.05, median 62.50, minimum 5.00, maximum 1,964.50, population standard deviation 179.92. Standard deviation measures how widely values vary around the mean. MEDIAN(line_value) describes lines and gives 22.00.

```
Median Order Value =
CALCULATE(
MEDIANX(VALUES(sales_lines[order_id]), [Total Revenue]),
sales_lines[status] = "completed"
)
```

Median order	Mean order	Largest order
62.50	**114.05**	**1,964.50**

Limits and common mistakes: use basic statistical functions

When another option fits

Report the mean when the average contribution matters. For a typical order in a skewed distribution, also inspect the median. STDEV.P describes the population, while STDEV.S estimates from a sample.

The mistake to diagnose

The fact contains several lines per order. A line-level statistic answers a different question. Of 2,771 completed orders, 2,082 are below the mean and 102 reach 500 or more.

Exam practice: use basic statistical functions

A few large orders raise average order value. The manager wants a typical completed order, with less sensitivity to those large values. Which calculation fits?

A MEDIAN(sales_lines[line_value]).

B MEDIANX over distinct completed order IDs, evaluating order revenue.

C MAX(sales_lines[line_value]).

D COUNTROWS(sales_lines).

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · use basic statistical functions

A This finds a typical line, not a typical order.

B The median uses order-grain observations and is less affected by extreme orders.

C The largest line is neither a typical order nor its total.

D A row count does not describe the middle order value.

Full worked sequence: D20 pp. 74–75 · Lab D19 / 02; D20 required study

Microsoft Learn · statistics

Closing stock adds across products but not across dates

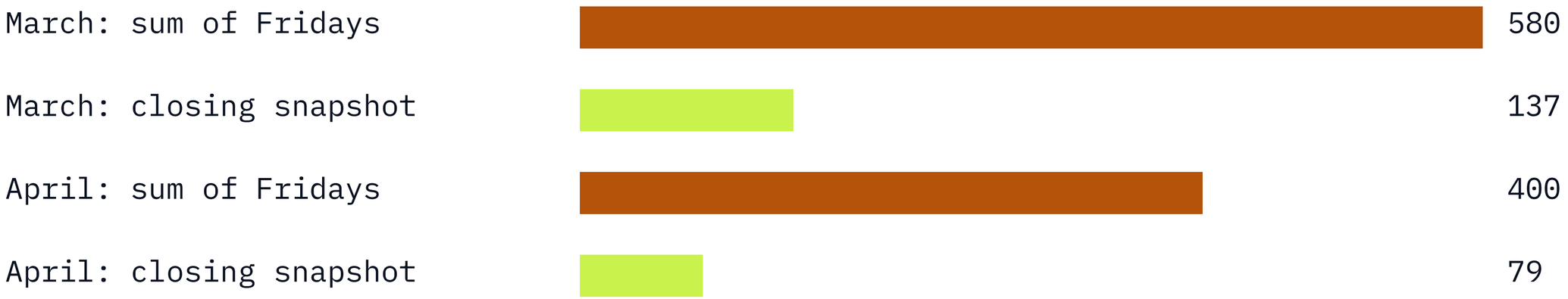
A semi-additive measure adds along some dimensions but needs another rule along others. Stock units add across products on the same date. Across dates, closing stock uses the last relevant snapshot.

Product	03-06	03-13	03-20	03-27	04-03	04-10	04-17	04-24
P-014	37	35	35	34	25	29	27	26
P-026	20	60	44	39	35	29	23	17
P-059	55	43	51	46	43	32	29	26
P-002	24	21	18	18	18	17	14	10

03-27 is the last March snapshot. Sum its four products: 34 + 39 + 46 + 18 = 137 units.

The monthly sum repeats stock held on earlier Fridays

March’s Friday snapshots sum to 580 units, but the closing snapshot has 137. April’s sum is 400, with 79 at close. Fleece jacket closes at 34 and 26 units.



Units on hand are balances. Adding repeated balances across dates overstates closing stock.

The last populated date differs from the calendar month end

March ends on Tuesday 31 March, with no snapshot. LASTDATE selects that empty day. LASTNONBLANK finds Friday 27 March. Stock on Hand then returns 137 units across the four products.

```
Stock on Hand =
CALCULATE(
SUM(inventory_snapshots[units_on_hand]),
LASTNONBLANK(dates[date],
CALCULATE(COUNTROWS(inventory_snapshots)))
)
```

```
CLOSINGBALANCEMONTH(SUM(inventory_snapshots[units_on_hand]), dates[date])
LASTDATE(dates[date]) selects 31 March in the March row.
```

Within March, the last populated date is 27 March. LASTDATE instead selects 31 March, which has no stock row.

Limits and common mistakes: create semi-additive measures

When another option fits

Sales revenue is additive across dates. Closing stock needs a stated balance convention. The stock file records every product on every Friday. Missing product snapshots require a separate carry-forward policy.

The mistake to diagnose

CLOSINGBALANCEMONTH evaluates at month end. With no record on that date, SUM is blank. AVERAGE across March’s product-Friday rows gives 36.25, which is neither total closing stock nor average weekly total.

Exam practice: create semi-additive measures

Four complete weekly stock snapshots are stored for each product in March. The report must show units held at the last recorded March snapshot, summed across products. Which rule fits?

A SUM units across every March snapshot.

B AVERAGE all product-Friday rows.

C Evaluate SUM at the last date in March with snapshot rows.

D Evaluate SUM only on 31 March even when it has no snapshot.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: C · create semi-additive measures

A This counts stock repeatedly and gives 580 units.

B This averages product-Friday rows and gives 36.25.

C The last populated date is 27 March, giving 137 units.

D There is no 31 March row, so the balance is blank.

Full worked sequence: D20 p. 71; guide pp. 55–60 · Lab D20 / 06 step 6

Microsoft Learn · stock

A quick measure generates ordinary DAX from your selections

A quick measure is a model measure whose DAX Power BI generates from a template. You select the calculation and fields, then inspect and edit the formula as needed.

Quick-measure selection	Value
Calculation	Year-to-date total
Base value	Completed Revenue
Date	dates[date]
DAX result type	Editable model measure

Illustrative selection layout. Inspect the DAX generated by your version, then compare it with Revenue YTD.

The quick YTD measure should match the handwritten measure

Choose Year-to-date total, Completed Revenue as the base value and dates[date] as the date. April 2026 should return 62,497.50, matching Revenue YTD. Inspect the generated DAX before relying on it.

```
Revenue YTD =
TOTALYTD([Completed Revenue], dates[date])
```

Generated quick measure	Handwritten check
62,497.50	**62,497.50**

Equivalent handwritten check. Generated syntax may differ. Select April 2026 and compare both measures.

Limits and common mistakes: create a measure by using quick measures

When another option fits

Write DAX directly when the template does not represent the business rule. The available controls differ between Desktop and the web, and between models. The generated expression needs the same correctness checks as handwritten DAX.

The mistake to diagnose

Picking the sparse sales date instead of the shared calendar can produce a different time context. Generated DAX may include extra checks or a longer expression. A template cannot repair relationships or date quality.

Exam practice: create a measure by using quick measures

You create a quick measure and need to adjust its business logic. What is the appropriate next action?

A Edit its generated DAX and check the result.

B Convert it to a calculated column.

C Rebuild the model because generated measures cannot change.

D Treat its result as correct because it came from a template.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: A · create a measure by using quick measures

A It is a normal model measure, so its DAX can be inspected and changed.

B A stored column would change when and how the calculation evaluates.

C Generated measures can be edited like other measures.

D Field choices and model assumptions can still be wrong.

Full worked sequence: D20 pp. 43, 76 · Lab D20 / 03 step 4

Microsoft Learn · quick

Calculated columns evaluate a formula for each current row

Row context identifies the current row during an iteration or calculated-column evaluation. A calculated column stores the result. RELATED uses a relationship to retrieve a value from the related row.

Order	Product	qty	price	Stored DAX Line Value
100142	Compression top	3	16.00	48.00

```
DAX Line Value = sales_lines[qty] * sales_lines[price]
Category = RELATED(products[category])
```

The current line supplies qty and price. RELATED follows product_id to retrieve its category.

A calculated table can store a category revenue summary

ADDCOLUMNS evaluates the measure for each category. Calling the measure turns the category row into a filter through context transition. The stored summary has six rows, with Outerwear 76,637.00.

```
Category Revenue =
ADDCOLUMNS(
VALUES(products[category]),
"Completed Revenue", [Completed Revenue]
)
```

Category	Stored Completed Revenue
Outerwear	76,637.00
Shoes	62,373.50
Bottoms	50,535.00

First 3 of 6 category rows

Limits and common mistakes: create calculated tables or columns

When another option fits

CALENDAR builds dates. A copy of dates can provide a second role. SUMMARIZECOLUMNS can group and calculate results. Stored summaries do not recalculate their values when a user moves a slicer.

The mistake to diagnose

RELATED requires a suitable current row and relationship. A bare RELATED in a measure has no current row. Measures inside an iterator can perform context transition, which differs from simply summing a column.

Exam practice: create calculated tables or columns

You need a category label on each sales line for grouping. The products dimension has a unique product ID and a category column. Which DAX calculated-column expression fits?

A RELATED(products[category]).

B SUM(products[category]).

C SELECTEDMEASURE().

D TOTALYTD(products[category], dates[date]).

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: A · create calculated tables or columns

A The current sales line identifies its related product row and category.

B SUM cannot aggregate text category labels.

C SELECTEDMEASURE belongs in a calculation item, not this column.

D Time intelligence does not retrieve a related text attribute.

Full worked sequence: D20 pp. 33, 67–70 · Lab D20 / 03 step 3 and 06 step 1

Microsoft Learn · tables

A calculation group reuses a transformation across measures

A calculation group contains named calculation items. Each item can use SELECTEDMEASURE as a placeholder for the explicit measure being evaluated. This avoids repeating the same time transformation for every measure.

April 2026 · completed	Current	YTD	Last year
Completed Revenue	20,237.00	62,497.50	12,791.00
Orders	137	539	103

The item name selects a transformation. SELECTEDMEASURE refers to the explicit measure in each Values cell.

Three calculation items apply to both revenue and orders

With April 2026 and completed selected, Current, YTD and Last year produce 20,237.00, 62,497.50, 12,791.00 for revenue, and 137, 539, 103 for Orders.

```
Current: SELECTEDMEASURE()
YTD: CALCULATE(SELECTEDMEASURE(), DATESYTD(dates[date]))
Last year: CALCULATE(
SELECTEDMEASURE(), SAMEPERIODLASTYEAR(dates[date]))
```

April 2026 · completed	Current	YTD	Last year
Completed Revenue	20,237.00	62,497.50	12,791.00
Orders	137	539	103

Limits and common mistakes: create calculation groups

When another option fits

Use a simple YTD measure if only one measure needs it. Create groups in Desktop's Model view. Enabling Discourage implicit measures means new dragged-column aggregations should be replaced with explicit measures.

The mistake to diagnose

Field parameters select fields. Calculation groups apply calculations to measures. A ratio or percentage may need exclusions or special handling, so a reusable item should not blindly transform every measure.

Exam practice: create calculation groups

Revenue, orders and units each need Current, YTD and Last year. Which feature lets you maintain the time transformations once and apply them to explicit measures?

A A calculation group using SELECTEDMEASURE.

B A calculated column for each report cell.

C A field parameter alone.

D A bidirectional relationship.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: A · create calculation groups

A Calculation items can transform whichever explicit measure is being evaluated.

B Stored row values cannot represent each visual’s changing filter context.

C Field parameters switch fields but do not supply reusable YTD transformations.

D Direction controls filters, not reusable DAX transformations.

Full worked sequence: D20 pp. 72–73 · Lab D20 / 06 step 3

Microsoft Learn · groups

Chapter 3: optimize model performance

Storage, visual timing and the cost of retaining detail

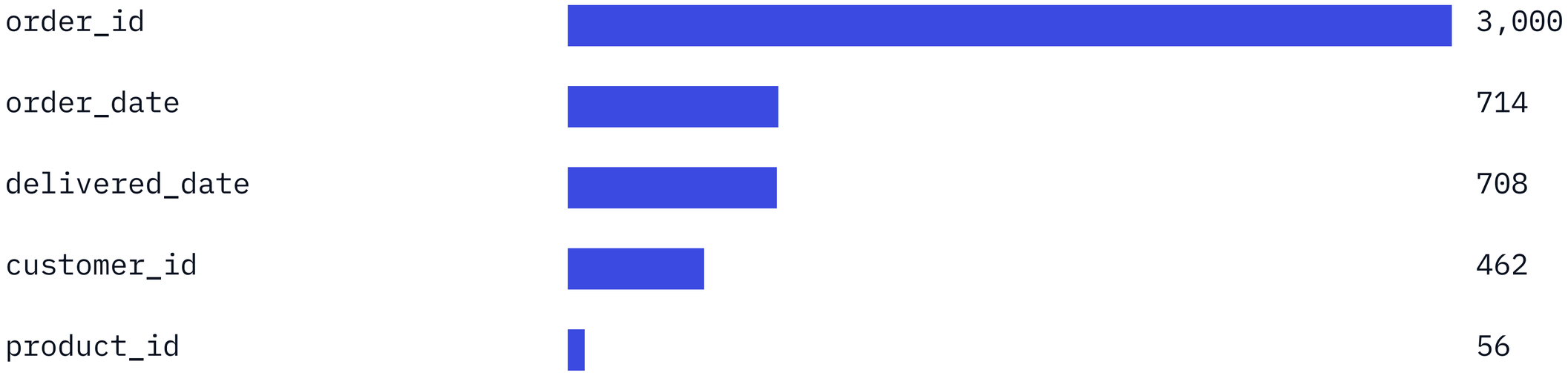
Design the model

Write calculations

Improve performance

Unused columns and excess detail increase model storage

Import models compress columns. More distinct values generally require larger dictionaries or encodings. Row count, data type and value distribution also matter. Distinct-value counts alone do not predict exact memory.



These are distinct-value counts, not measured bytes. Check actual storage before estimating savings.

Remove unused fields before load and hide required keys

Keep product_id for the relationship, then hide it if report authors do not need it. Remove unused high-detail identifiers and timestamps in Power Query. Hiding a field leaves its data in the model.

Field	Action	Reason
product_id	Keep and hide	Required relationship key
Unused unique text	Remove before load	No downstream dependency
status	Keep	Cancellation analysis

Removing and hiding have different effects on storage. Check the report’s questions before deleting data.

Limits and common mistakes: identify and remove unnecessary rows and columns

When another option fits

Retain the rows and fields needed by the agreed questions, relationships and security. Check downstream dependencies before removing a field. Reducing a numeric precision can help only when the business accepts the change.

The mistake to diagnose

Removing cancelled and pending rows removes 565 lines. Revenue becomes 316,023.50 instead of 340,370.00. The cancellation rate, 212 of 3,000 orders or 7.1%, can no longer be calculated from the remaining data.

Exam practice: identify and remove unnecessary rows and columns

An imported text column has a different value on nearly every row. It serves no visual, relationship, measure or security rule. How should you reduce its storage?

A Hide it in Report view.

B Remove it before loading into the model.

C Change the report theme.

D Set its display folder.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · identify and remove unnecessary rows and columns

A Hiding changes visibility but keeps the stored column.

B Removing it before load eliminates its model storage.

C A theme changes report appearance, not imported data storage.

D Display folders organise fields without removing data.

Full worked sequence: D18 pp. 82–89 · Lab D18 / 04

Microsoft Learn · size

Performance Analyzer separates query time from visual display

Performance Analyzer records report visual durations. DAX query is time for a visual’s request. Visual display is drawing time. Other includes preparation and waiting. A duration is evidence to investigate, not a diagnosis by itself.

Visual · illustrative ms	DAX query	Visual display	Other
Revenue card	620	20	40
Detailed scatter	80	650	40
Waiting visual	60	30	700

ILLUSTRATIVE timings, not measurements. Investigate the longest component before choosing a change.

DAX query view returns the data behind a visual

EVALUATE returns a table expression. SUMMARIZECOLUMNS groups results by category. The first row after descending revenue sort is Outerwear, 76,637.00. DEFINE MEASURE can test query-scoped logic without saving it to the model.

```
DEFINE MEASURE sales_lines[Test Revenue] = [Completed Revenue]
EVALUATE
SUMMARIZECOLUMNS(
products[category], "Revenue", [Test Revenue])
ORDER BY [Revenue] DESC
```

Category	Revenue
Outerwear	76,637.00
Shoes	62,373.50
Bottoms	50,535.00

First 3 of 6 category rows

Desktop and the service expose different entry points

Desktop uses Optimize → Performance analyzer and the DAX query view icon. The service report editor uses View → Performance analyzer. Open Write DAX queries on the semantic model to run a copied query.

Tool	Desktop	Service
Performance Analyzer	Optimize ribbon	Edit report → View
Run visual query	Run in DAX query view	Copy query, then model query view
DAX query view	Left-side view icon	Write DAX queries on model
Save query tabs	Saved with file	Discarded on close

Web DAX queries need model write permission and the workspace editing setting.

Limits and common mistakes: use Performance Analyzer and DAX query view

When another option fits

A slow DAX request needs investigation of its measure, relationships, query and source. Long visual display suggests inspecting the visual’s complexity. Retest under comparable filters and cache conditions before claiming an improvement.

The mistake to diagnose

Performance Analyzer times report visuals, not Power Query refresh. Use Query Diagnostics for query transformations. Other time can include waiting behind other visuals, so it is not a direct measure of DAX cost.

Exam practice: use Performance Analyzer and DAX query view

Performance Analyzer shows a long DAX query duration and a short Visual display duration for a matrix. Which investigation should come first?

A Change the matrix’s colours.

B Copy its query and inspect measures and filter paths in DAX query view.

C Remove every relationship immediately.

D Use Performance Analyzer to measure a Power Query refresh.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: B · use Performance Analyzer and DAX query view

A Colour changes do not address the long query duration.

B The query and model are the first places to investigate this timing pattern.

C A timing pattern alone does not justify removing required relationships.

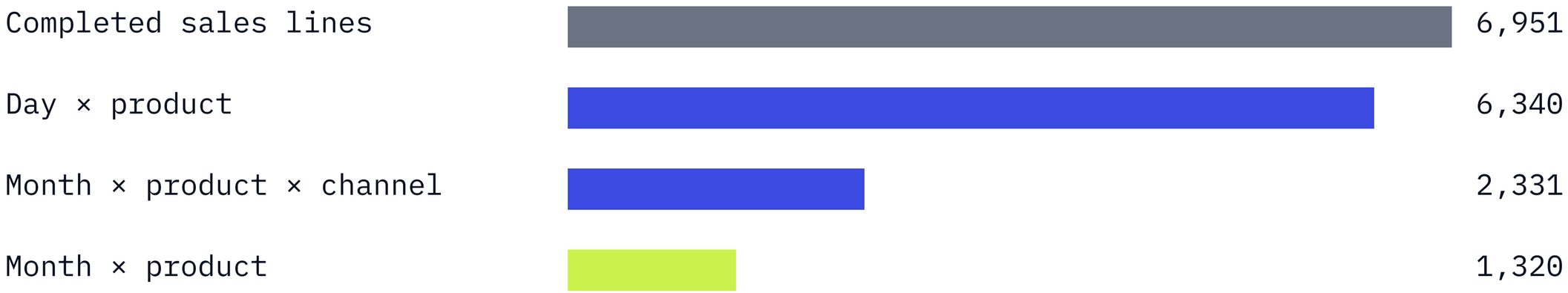
D This tool measures report visual interactions, not Power Query refresh.

Full worked sequence: Guide pp. 82–87 · Lab D20 / 06 step 5

Microsoft Learn · analyzer

A coarser fact grain stores fewer rows and less detail

Grain says what one fact row represents. Reducing granularity summarises detailed rows to a coarser level. The design must preserve the questions users need and the dimensions used to filter them.



Month × product reduces rows by about 81%. The report can no longer recover individual orders or sales days.

Monthly product totals preserve revenue but lose order detail

Completed lines: 6,951. Day × product: 6,340 rows. Month × product × channel: 2,331. Month × product: 1,320. The last table keeps 316,023.50, April 20,237.00 and Outerwear 76,637.00.

Month × product keeps	It cannot recover
Completed revenue 316,023.50	Orders 2,771
April revenue 20,237.00	Customers 452
Outerwear 76,637.00	Daily and delivery-date detail

Revenue totals add across the retained groups. Distinct orders and customers need detail that this table discards.

Exam practice: reduce granularity

Managers need only completed revenue by month and product. They do not need order counts, customer detail, daily analysis or delivery dates. Which grain gives the largest reduction among these options while preserving their question?

A The original sales-line grain.

B Day × product.

C Month × product.

D Year only.

Choose one answer before turning the page. These are original practice questions, not Microsoft exam items.

Answer: C · reduce granularity

A It preserves detail the stated report does not need.

B It retains daily detail and reduces this fixture by only about 9%.

C It retains the required month and product dimensions, with 1,320 rows.

D It discards the month and product breakdowns the managers require.

Full worked sequence: D18 pp. 29–31; guide pp. 88–91 · Lab D18 grain discussion; guide example

Microsoft Learn · size

Mixed review: scenarios 1–3

Delivery-date YTD

A manager wants YTD revenue by delivery date while the report retains its normal order-date default.

Stock with no month-end snapshot

March ends on a Tuesday but the file records only Fridays. The closing balance is blank.

A category slicer and a stored summary

A calculated category table keeps Outerwear at 76,637.00 after selecting online. A measure shows 25,315.50.

Write the model change or calculation rule, then name the exam skills that justify it.

Mixed review: scenarios 4–6

A slow matrix with Both direction

A matrix is slow after broad direction changes, but no timing evidence has been collected.

A monthly summary and customer counts

A proposed monthly product table has revenue and a distinct customer count for every group. Can the grand total sum those counts?

Years displayed as a large number

A card sums the calendar year column. Hiding it makes the card harder to build but does not change the stored data.

Write the model change or calculation rule, then name the exam skills that justify it.

Mixed review answers: scenarios 1–3

Delivery-date YTD

Use a measure that activates the existing delivery-date relationship and applies the YTD date window. Keep the default order-date relationship. Re-read 1.2 and 2.3.

Stock with no month-end snapshot

Select the last populated snapshot date inside the period. In this complete fixture, 27 March returns 137 units. Re-read 2.5.

A category slicer and a stored summary

The table stores refresh-time values. Use the measure when the value must respond to the selection. Re-read 1.5 and 2.7.

Mixed review answers: scenarios 4–6

A slow matrix with Both direction

Record a comparable interaction, inspect query and display durations, then test required filter paths. Do not assume Both is the sole cause. Re-read 1.3 and 3.2.

A monthly summary and customer counts

Customers can occur in several groups. Keep suitable detail or another valid distinct-count design. Re-read 3.3.

Years displayed as a large number

Set the default summarisation appropriately and use year as a label. A year sum has no useful business meaning here. Re-read 1.1 and 3.1.

Revision index: guide, class sequence and practice

1.1 · guide 6 · D18 pp. 47, 82–85

1.2 · guide 12 · D18 pp. 74–81

1.3 · guide 17 · D18 pp. 53–68

1.4 · guide 22 · D18 pp. 42–50; D20 pp. 8–17

1.5 · guide 27 · D19 pp. 11–13; D20 pp. 67–70

2.1 · guide 33 · D19 pp. 15–28

2.2 · guide 38 · D19 pp. 38–64; D20 pp. 38–40, 66

2.3 · guide 44 · D20 pp. 8–29, 63–64

2.4 · guide 50 · D20 pp. 74–75

2.5 · guide 55 · D20 p. 71; guide pp. 55–60

2.6 · guide 61 · D20 pp. 43, 76

2.7 · guide 66 · D20 pp. 33, 67–70

2.8 · guide 71 · D20 pp. 72–73

3.1 · guide 77 · D18 pp. 82–89

3.2 · guide 82 · Guide pp. 82–87

3.3 · guide 88 · D18 pp. 29–31; guide pp. 88–91

Each answer page links its Microsoft Learn source and names the class lab. Use the PDF bookmarks to revisit a skill.

Before the practice assessment

Use Microsoft’s practice assessment to check your reasoning. Review mistakes by skill, then explain a worked example without reading its formula. This guide covers one of the four exam domains.

Microsoft PL-300 study guide

Current Microsoft practice assessment

DAX and model cheat sheet

Inventory snapshot practice

Companion guides for Prepare, Visualize and Manage are planned separately. Use Microsoft Learn for those domains meanwhile.